Technical Report: Exploratory Data Analysis on Marketing Campaign Dataset

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Project: Exploratory Data Analysis for Marketing Campaign Optimization
Tools Used: Python, Pandas, Seaborn, Matplotlib, NumPy

Table of Contents

1. Introduction

This report conducts an in-depth exploratory data analysis (EDA) on a U.S.-based marketing campaign dataset containing over 200,000 campaign instances. It aims to uncover patterns in campaign performance, user engagement, and regional effectiveness across different marketing channels such as YouTube, Instagram, and Google Ads.

2. Objectives

The primary objectives of this EDA project are:

- To analyze key marketing performance metrics such as ROI, CTR, CPC, and Conversion Rate.
- To evaluate campaign success across cities and channels.
- To discover trends and relationships that can inform future campaign planning and execution.
- To recommend actionable marketing strategies based on data-driven evidence.

3. Data and Methodology

3.1 Data Sources

The dataset was sourced from internal marketing analytics platforms and includes campaigns run between **January and December 2021** across five major U.S. cities:

- Chicago
- New York
- Los Angeles
- Miami
- Houston

3.2 Tools and Technologies

- **Python** for data manipulation and analysis
- **Pandas** for data wrangling
- **Seaborn & Matplotlib** for visualization
- **NumPy** for metric engineering and statistics

3.3 Analytical Workflow

1. Data loading and initial inspection
2. Cleaning and formatting (e.g., datetime conversion, NA handling)
3. Engineering new metrics:
    - CTR = Clicks / Impressions
    - CPC = Acquisition_Cost / Clicks
4. Outlier detection and treatment using standard deviation thresholds
5. Visualization and interpretation of trends

4. Results

4.1 ROI by Marketing Channel

YouTube exhibited the **highest average ROI**, outperforming Display and Email campaigns significantly. It consistently yielded returns above the median benchmark across all five cities.

4.2 CTR vs. Conversion Rate

A moderate **positive correlation** was found between Click-Through Rate and Conversion Rate. However, high CTR values **did not always result in high conversions**, emphasizing the need for better **landing page experiences and content relevance**.

4.3 Monthly Click Trends

The second quarter (April to June) demonstrated the **highest user click volume**, suggesting seasonal responsiveness. This peak aligns with increased online engagement in spring.

4.4 ROI Heatmap: Campaign Type by City

- **Influencer campaigns** had superior ROI in **Los Angeles and Miami**, likely due to lifestyle-centric consumer behavior.
- **Email campaigns** underperformed in most cities except Chicago.
- **Display ads** had consistent but lower ROI across all cities.

5. Discussion

5.1 Channel Effectiveness

YouTube’s high ROI confirms the value of **video-based, immersive content**. It is especially effective in capturing attention and driving conversions.

5.2 Seasonal and Geographic Insights

- Q2 campaigns benefited from favorable **seasonal user behavior**.
- Coastal cities like Miami and LA responded better to **Influencer-led promotions**, reflecting lifestyle alignment.

5.3 Correlation Observations

CTR is **not a standalone success indicator**. Campaigns need **holistic design**—including content, landing pages, and offers—to convert clicks into actions.

5.4 Segmentation Opportunities

Behavioral and demographic trends suggest that **Men aged 25–34** and **Women aged 35–44** responded most favorably to targeted messaging, especially on video platforms.

6. Recommendations

6.1 Channel Prioritization

- Increase investment in **YouTube and Influencer Marketing** channels.
- Reduce reliance on low-performing Email campaigns.

6.2 Timing Strategy

- Schedule high-budget campaigns in **Q2** to align with elevated user activity.
- Conduct A/B testing for off-peak months.

6.3 UX and Conversion Optimization

- Improve **landing page content, load times**, and **call-to-action clarity**.
- Match ad messaging with post-click user experience.

6.4 Target Audience Refinement

- Personalize ads for **Men (25–34)** and **Women (35–44)** segments.
- Leverage cross-channel insights to deepen segmentation.

7. Deliverables

File Name	Description
eda_marketing_clean.csv	Cleaned and enhanced dataset for analysis
marketing_eda_visuals.png	Key visualizations (e.g., ROI heatmap, trends)
Technical Report (this document)	Full EDA workflow and insights documentation

8. Conclusion

This technical report highlights the power of exploratory analysis in understanding campaign effectiveness across digital marketing channels. Key insights—including high ROI from YouTube and Influencer strategies, seasonal performance spikes, and demographic targeting—can significantly shape future marketing decisions. With clear recommendations in place, the organization is well-positioned to refine strategy, improve ROI, and scale customer engagement.

9. References

- Python & Seaborn Documentation
- https://www.kaggle.com/datasets/manishabhatt22/marketing-campaign-performance-dataset